

ORION

PART 05

AI Orchestration Layer

AI agents, decision support, planning and coordination

Technical Architecture Series • Beginner → Professional

How to use this volume

This document is one independent volume in the twelve-part ORION technical series. Beginners should read from the start; experienced readers can jump directly to the relevant architecture sections.

ORION is a secure communications, infrastructure-coordination, emergency-response, and decision-support architecture. High-impact weapons-control functionality is deliberately outside this design.

Purpose

Provide analysis, prediction, planning, and decision support through specialized AI services rather than one unrestricted agent.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Agent model

Telemetry/events → situation awareness → specialized models → policy guard → human approval for high-impact actions → authorized execution.

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Specialized agents

Network Analyst; Security Analyst; Incident Analyst; Capacity Planner; Simulation Agent; Coordinator.

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AI inputs

Network telemetry, service health, incident reports, approved external feeds, configuration, historical data, and simulation state.

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AI outputs

Predictions, summaries, risk scores, recommendations, scenario comparisons, and explanations.

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AI safety

Models should not hold unrestricted credentials. Tools are mediated by APIs and independent policy checks. Model text is never itself authorization.

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Evaluation

Accuracy, false positives/negatives, calibration, latency, robustness, prompt-injection resistance, data-poisoning resistance, and operator acceptance.

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Failure behavior

If AI is unavailable or confidence is low, fall back to deterministic policies and human workflows.

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Architecture Review Checklist

- Requirements and assumptions are explicit.
- Component ownership and boundaries are documented.
- Normal, degraded, and failure states are defined.
- Security and privacy are architectural requirements.
- Capacity and scaling assumptions are measurable.
- Logs, metrics, traces, and audit events cover critical flows.
- Dependencies have explicit contracts and failure behavior.
- Important decisions record alternatives and trade-offs.
- Testing covers expected behavior and controlled failures.
- Operational runbooks exist for important incidents.

Key Takeaways

AI Orchestration Layer should be implemented as a bounded layer of ORION, with explicit interfaces to the rest of the platform.

The goal is not maximum complexity. The goal is measurable resilience, security, maintainability, interoperability, and operational usefulness.

Glossary

- API — Application Programming Interface used for software-to-software communication.
- Edge — Computing placed near users or physical systems.
- Event — A record that something happened.
- Failover — Switching to a redundant component after failure.
- Idempotency — Repeating an operation does not create an unintended additional effect.
- NTN — Non-terrestrial network, commonly involving satellite infrastructure.
- Policy — A rule that determines what is allowed under stated conditions.
- RPO — Recovery Point Objective; acceptable data-loss window.
- RTO — Recovery Time Objective; target recovery time.
- Zero trust — Continuous verification instead of implicit trust based on network location.